The Anomalous Magnetic Moment of the Muon in Unified Mechanics

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2026

Abstract

The Unified Mechanics axiom $c^2 = c + 1$ derives the fine structure constant α_{em} via the Fibonacci cycle sum $1/\alpha = \varphi^{10} + \varphi^5 + \varphi^2 + \varphi^{-2} = 137.082$, deviating from the CODATA value by +0.034% ($0.011\,\varepsilon_{\text{floor}}$). This pinning of α_{em} fixes all QED contributions to the muon anomalous magnetic moment a_μ to sub-floor precision. A second UM contribution addresses the hadronic vacuum polarisation (HVP): confinement geometry imposes a correction of order $\varepsilon_{\text{floor}} = r^3$ on each hadronic loop, yielding $\varepsilon_{\text{floor}} \times a_\mu^{\text{HVP}} \approx 2.05 \times 10^{-9}$. The observed discrepancy between experiment and SM theory is $\Delta a_\mu = 2.51 \times 10^{-9}$ (4.2σ), an 18.5% residual from the UM leading-order estimate — consistent with the framework resolution, which requires the full UM-QCD dispersion integral to reach sub-floor precision.

1 Foundations

The single recursion $c^2 = c + 1$ and its immediate consequences (Paper 1):

$$c^2 = c + 1 \quad \Rightarrow \quad \varphi = \tfrac{1+\sqrt{5}}{2}, \quad r = \tfrac{1}{2\varphi} \approx 0.30902. \tag{1}$$

The three channel weights and the braiding floor:

$$W_L = (1 - r)^2 \approx 0.4775, \tag{2}$$
$$W_B = 2r(1 - r) \approx 0.4271, \tag{3}$$
$$W_M = r^2 \approx 0.0955, \tag{4}$$
$$\varepsilon_{\text{floor}} = r^3 \approx 0.02951. \tag{5}$$

The anomalous magnetic moment $a_\mu \equiv (g_\mu - 2)/2$ is measured to parts-per-billion precision and computed in the SM from QED, EW, and hadronic contributions. The current experimental situation is:

$$a_\mu^{\rm exp} = 116\,592\,061(41) \times 10^{-11} \quad \text{(FNAL 2023)}, \tag{6}$$

$$a_\mu^{\rm SM} = 116\,591\,810(43) \times 10^{-11} \quad \text{(dispersive HVP, White Paper)}, \tag{7}$$

$$\Delta a_\mu = 2.51 \times 10^{-9} \quad (4.2\sigma). \tag{8}$$

UM enters this calculation at two points: through the value of $\alpha_{\rm em}$ that feeds into every QED diagram, and through a confinement-geometry correction to the hadronic loops.

2 The Fine Structure Constant in UM

2.1 Derivation via Fibonacci cycle sum

Paper 6 of the UM corpus identifies the Fibonacci cycle sum

$$\frac{1}{\alpha_{\rm em}} = \varphi^{10} + \varphi^5 + \varphi^2 + \varphi^{-2} = 137.082. \tag{9}$$

The exponents $\{10, 5, 2, -2\}$ are the cycle-winding numbers of the electromagnetic zero-mode on the four independent cycle types of the G_2 compactification manifold (Paper 6; see also the companion paper on G_2 cycle geometry). The sum arises because the photon couples simultaneously to all four cycle types; the coupling strength is the inverse of the total winding sum.

2.2 Numerical result

$$\varphi^{10} = 122.992, \tag{10}$$

$$\varphi^5 = 11.090, \tag{11}$$

$$\varphi^2 = 2.618, \tag{12}$$

$$\varphi^{-2} = 0.382, \tag{13}$$

$$\text{Sum} = 137.082. \tag{14}$$

The CODATA 2018 value is $1/\alpha = 137.036$.

$$\text{Residual} = \frac{137.082 - 137.036}{137.036} = +0.034\% = 0.011\,\varepsilon_{\rm floor}. \tag{15}$$

This is the most precise result in the UM corpus. The 0.034% deviation is sub-floor — below the minimum resolvable shift of one $\varepsilon_{\rm floor}$.

2.3 Consequence for a_μ

The QED contribution to a_μ (Schwinger term plus radiative corrections through three loops) is:

$$a_\mu^{\mathrm{QED}} \approx 116\,584\,718 \times 10^{-11}. \tag{16}$$

Since α_{em} is fixed to $0.011\,\varepsilon_{\mathrm{floor}}$ precision by UM, all QED contributions to a_μ are correspondingly pinned. No adjustment to a_μ^{QED} is required by UM. The entire residual discrepancy Δa_μ therefore lies in the hadronic sector.

3 The Hadronic Vacuum Polarisation

3.1 Confinement geometry correction

In UM, the proton/electron mass ratio is derived as $m_p/m_e = \varphi^{15}(1+r)(1+r^3)$ (electroweak/baryon paper). The factor $(1+r)$ encodes the confinement geometry: a quark traversing the boundary channel acquires a factor $(1+r)$ relative to the free propagator. The same geometric factor applies to any hadronic loop in the vacuum polarisation.

For the HVP contribution a_μ^{HVP}, each hadronic loop closure involves one boundary-channel traversal with the confinement correction. At leading order in $\varepsilon_{\mathrm{floor}}$, the fractional shift imposed by UM is:

$$\frac{\delta a_\mu^{\mathrm{HVP}}}{a_\mu^{\mathrm{HVP}}} = \varepsilon_{\mathrm{floor}} = r^3. \tag{17}$$

3.2 Numerical result: dispersive approach

The dispersive (data-driven) HVP estimate is $a_\mu^{\mathrm{HVP,disp}} = 6931 \times 10^{-11}$ (White Paper 2020). The UM leading-order correction is:

$$\begin{aligned}
\delta a_\mu^{\mathrm{UM,disp}} &= \varepsilon_{\mathrm{floor}} \times a_\mu^{\mathrm{HVP,disp}} \\
&= r^3 \times 6931 \times 10^{-11} \\
&= 0.02951 \times 6931 \times 10^{-11} \\
&\approx 2.045 \times 10^{-9}.
\end{aligned} \tag{18}$$

The observed discrepancy $\Delta a_\mu = 2.51 \times 10^{-9}$.

$$\text{Residual} = \frac{2.045 - 2.51}{2.51} = -18.5\%. \tag{19}$$

This 18.5% residual is larger than $\varepsilon_{\mathrm{floor}}$, which identifies the present estimate as a leading-order result. The full computation requires integrating the dispersion relation with the UM-derived $\alpha_s = r^2(1 + r - r^2)$ (Paper 5), which modifies the hadronic spectral function at the UM scale. That integral is identified as the next calculation in the UM particle-physics programme. The order-of-magnitude agreement — both estimates in the range 2–3×10^{-9} — is notable given the simplicity of the leading-order formula.

4 The Dispersive vs. Lattice Tension

The BMW 2020 lattice QCD calculation gives $a_\mu^{\mathrm{HVP,lat}} = 7075 \times 10^{-11}$, higher than the dispersive estimate by 144×10^{-11} (+2.1%). If the lattice value is adopted, the SM discrepancy falls below 2σ.

This tension is a UM-resolvable question. The dispersive integral evaluates the $e^+e^- \to$ hadrons cross section measured in the continuum, where confinement effects are not directly observable. The lattice calculation is performed in the confined phase, where boundary-channel geometry is intrinsic. The UM confinement correction $(1+r)$ therefore applies differently to the two approaches:

- Dispersive: the spectral function is measured on free hadronic final states; the $(1 + r)$ boundary correction must be added as a post hoc shift.
- Lattice: the simulation is performed in the confined phase; the $(1+r)$ geometry is incorporated into the quark propagators automatically.

Quantitatively, applying $(1 + r)$ to the dispersive value gives:

$$a_\mu^{\mathrm{HVP,disp}} \times (1 + r) = 6931 \times 10^{-11} \times 1.30902 = 9073 \times 10^{-11}. \tag{20}$$

This overshoots the lattice value, indicating that $(1 + r)$ is the direction of the correction but that only a fraction of the boundary traversals are affected per hadronic loop. The fraction is of order $r^2 = W_M$, which gives $\delta a_\mu^{\mathrm{HVP}} \sim W_M \times r \times a_\mu^{\mathrm{HVP}} = r^3 \times a_\mu^{\mathrm{HVP}}$, recovering equation (**??**). The dispersive-lattice tension is therefore consistent with the UM framework at the level of $\varepsilon_{\mathrm{floor}}$, and the lattice calculation gives the more reliable baseline for the UM estimate.

For completeness, the UM leading-order correction to the lattice value:

$$\delta a_\mu^{\mathrm{UM,lat}} = r^3 \times 7075 \times 10^{-11} \approx 2.088 \times 10^{-9}. \tag{21}$$

The residual against $\Delta a_\mu = 2.51 \times 10^{-9}$ is -16.8%, marginally better than the dispersive estimate.

5 Discussion

The UM treatment of a_μ has two distinct components at different levels of precision.

The $\alpha_{\mathbf{em}}$ pinning is at sub-floor precision ($0.011\,\varepsilon_{\text{floor}}$). The QED contributions to a_μ are fixed to this precision by UM with no free parameters. This is the stronger result.

The HVP correction is an order-of-magnitude estimate at leading order in $\varepsilon_{\text{floor}}$. The formula $\varepsilon_{\text{floor}} \times a_\mu^{\text{HVP}}$ gives 2.0–2.1×10^{-9}, within 18% of the observed 2.51×10^{-9}. The residual is 6–7 $\varepsilon_{\text{floor}}$, which indicates a next-order term is needed. That term requires knowledge of the hadronic spectral function weighted by the UM-derived α_s. Once $\alpha_s(M_Z) = r^2(1 + r - r^2) = 0.1159$ is used in the running coupling, the dispersion integral can be re-evaluated.

Observable	UM expression	UM value	Observed	Error
$1/\alpha_{\text{em}}$	$\varphi^{10}+\varphi^5+\varphi^2+\varphi^{-2}$	137.082	137.036	$+0.034\%$ ($0.011\,\varepsilon_f$)
$\delta a_\mu^{\text{UM,disp}}$	$r^3 \times 6931 \times 10^{-11}$	2.045×10^{-9}	2.51×10^{-9}	-18.5%
$\delta a_\mu^{\text{UM,lat}}$	$r^3 \times 7075 \times 10^{-11}$	2.088×10^{-9}	2.51×10^{-9}	-16.8%

The path to a sub-floor a_μ result requires:

1. Using $\alpha_s^{\text{UM}} = r^2(1+r-r^2)$ in the hadronic running, re-computing the dispersive integral.
2. Implementing the full boundary-channel propagator correction in the lattice hadronic vacuum polarisation.
3. Resolving the dispersive-lattice tension, which UM frames as a difference in confinement-geometry incorporation.

The α_{em} pinning alone is a well-posed and precise UM result. The HVP calculation is in progress.